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https://cosmickb.org

Identification of curatable papers through an AI pipeline in COSMIC

Introduction

COSMIC, the Catalogue of Somatic Mutations in Cancer (Sondka et al 2024), is the world's largest source of expert manually curated somatic mutation information relating to human cancers. Data is curated from many sources including whole genome studies, large next generation sequencing panels and case reports from many different journals.

The first step in data curation is the identification of relevant papers from the literature. In COSMIC we focus searches on known cancer genes together with more focussed searches for tumour types. However, with hundreds of papers published in the field of cancer research every week, identifying those that contain somatic mutation data at the patient or sample level is a formidable task.

AI strategy

We decided to test Artificial Intelligence (AI) models to analyse papers returned from PubMed searches and select those that contain curatable data. A number of AI models were tested and trained using a large corpus of historical data spanning 20 years of papers tagged in COSMIC as either 'rejected' or 'curated'. Microsoft's PubMedBERT was found to give the best predictions and has been incorporated into a pipeline to run weekly PubMed searches and return a report of papers categorised as either 'curatable' or 'rejectable'. Subsequent testing has indicated that the predictions are over 85% accurate. We have also applied the pipeline to triage papers from specific tumour-type searches for more focussed COSMIC curation.

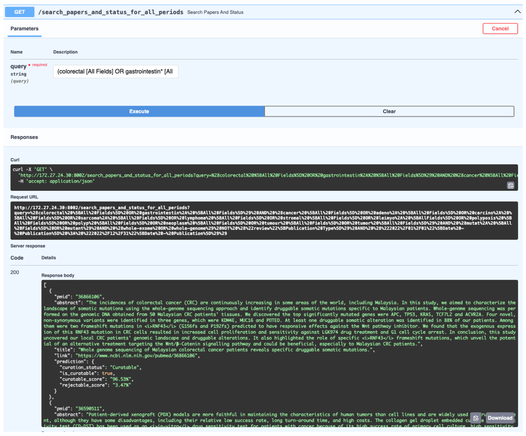
The challenge	Time-Consuming Analysis: Curators spend several hours analysing and identifying relevant research papers each week published via PubMed
The training set	Corpus of over 21,000 papers collected over 20 years classified as curated or rejected – provides positive (curatable) and negative (rejectable) data sets
The model	Four models from Hugging Face were tested: PubMedBERT, BioBERT, Mistral and Llama
The outcome	Weekly papers returned from the PubMed searches classified as either curatable or rejectable using the PubMed Bert model

Training the model

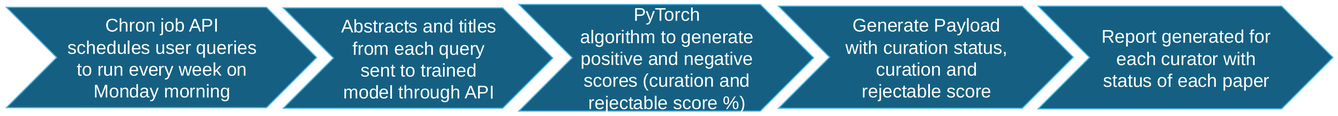
AI in operation- tumour-type focus search

AI applied to a colorectal cancer focus.
An API interface to the AI model was used to search PubMed for papers focused on colorectal cancers between 2019 and 2024. This was also divided into:
1. Large scale: whole-genome and whole-exome screens
2. Small scale: papers including targeted screens and case studies.

	Total number of papers retrieved from PubMed	Number assigned rejectable by AI	Number assigned curatable by AI
1. Large scale	607	301	306
2. Small scale	13,444	8895	4549



AI in operation- weekly PubMed searches



Query	User	Title	PubmedID	AI Curation Status	AI Curation Score	AI Rejectable Score
'MYB' (MYB [All Fields] OR C_MYB [All Fields]) AND (mutat* [All Fields] OR mutant [All Fields])	Rachel	The synergistic effect of c-Myb hyperactivation and Pu.1 deficiency induces Pelger-Huët anomaly and promotes sAML.	40020188	Curatable	83.46%	16.54%
'MYB' (MYB [All Fields] OR C_MYB [All Fields]) AND (mutat* [All Fields] OR mutant [All Fields])	Rachel	Phenotype of sickle cell disease. Correlation of haplotypes and polymorphisms in cluster β, BCL11A, and HBS1L-MYB. Pilot study.	40012971	Can be rejected	0.38%	99.62%

Accuracy of the AI predictions:
The AI assignment for 405 papers were manually checked.

	Correctly assigned as curatable	Incorrectly assigned as curatable	Correctly assigned as rejectable	Incorrectly assigned rejectable
Number of papers	149	25	219	12
% of papers	85.63	14.37	94.81	6.90

Conclusions and future work

- AI pre-screening of papers saves many hours of work allowing curators to focus on key curation tasks.
- The landscape of curatable cancer data can be more easily monitored and used to set priorities for curation.
- Future work will incorporate the AI searches into our curation pipeline to provide a user interface for searching and cataloguing screened papers.
- Further refinement of the model through analysis, feedback and re-training will reduce the numbers of false positives and negatives and address any bias in the process.
- False assignment is often due to missing data and particularly common for large scale sequencing where the full data is often omitted (Holmes et al (2024). Screening could be improved by including supplementary data and full text, including tables, in the training set.

References

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